PRISHA BANSAL

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PROFESSIONAL SUMMARY

Cybersecurity undergraduate interested in IoT, Embedded, and Autonomous Robotic and Drone Systems Security. Worked on real-time telemetry anomaly detection, UAV communication security (ROS 2/PX4, DDS), and threat-surface analysis. Also worked in building high-reliability detection pipelines achieving 100% attack detection with sub-second latency, alongside active research into emerging threat vectors in connected systems and applied AI architectures.

TECHNICAL SKILLS

- **IoT & Embedded Security:** Telemetry Spoofing & Anomaly Detection, Protocol Analysis (ROS 2, DDS/Micro XRCE-DDS, MAVLink), Firmware & Attack Emulation Basics, Network Anomaly Detection
- **Autonomous Systems & Robotics:** PX4 Autopilot, Gazebo SITL, MAVSDK, ROS 2, OpenCV
- **Programming Languages:** Python, C/C++, C#, SQL, Java, HTML/CSS
- **Security & Forensics:** C2 Architecture, Network Breach Analysis, MITRE CALDERA, Applied AI Security (RAG Indirect Prompt Injection)
- **Tools & Platforms:** Linux (Ubuntu/Kali), Git, Docker, Scikit-learn, NumPy, Pandas, Matplotlib, Wireshark

WORK EXPERIENCE

Research Intern | Helicopter Lab, IIT Kanpur

May 2026 – June 2026 | Kanpur, India

- Created a simulation on autonomous unmanned aerial vehicle (UAV) systems and telemetry integrity under Prof. Abhishek.
- Built an autonomous vision-guided tracking system integrating PX4 SITL, ROS 2, and Gazebo, validating camera-bridge telemetry streams and MAVSDK offboard command execution.

Security Engineering Intern | Treacle Technologies (IIT Kanpur)

February 2026 – May 2026 | Remote

- Analyzed embedded network and infrastructure threat surfaces under Mr. Raj Kumar.
- Assisted in assessing defensive security controls, testing automated monitoring workflows, and performing vulnerability analysis across connected endpoints.

PROJECTS & RESEARCH

UAV Telemetry Security Monitor | *ROS 2, PX4 SITL, Gazebo, Python*

- Engineered a real-time ROS 2 telemetry anomaly detection pipeline for autonomous connected UAVs operating in PX4/Gazebo SITL.
- Modeled empirical baseline flight profiles across 5 multi-phase missions using 99th-percentile velocity (6.59 m/s) and climb-rate thresholds.
- Detected 100% of telemetry spoofing attacks (5/5 attack runs) with sub-second latency and an ultra-low 0.25% false positive rate across 18,140+ telemetry data points.

Security Analysis of Indirect Prompt Injection in RAG Architectures | *Research Paper (In Progress)*

- Evaluated multi-layer prompt injection defenses under RAG pipelines using the BIPIA benchmark, isolating a 56% detection-rate drop caused by whole-document scoring dilution.
- Identifying granularity mismatch as a primary failure mode in RAG defense architectures.

Vision-Based Autonomous Drone Tracking System | *ROS 2, PX4, Gazebo, MAVSDK, OpenCV*

- Engineered a vision-based autonomous target tracking pipeline using PX4 SITL, Gazebo, ROS 2, and MAVSDK offboard control.
- Implemented real-time OpenCV HSV color segmentation and ROS 2 camera-bridge telemetry nodes to dynamically regulate yaw alignment and closed-loop distance control.

Network Breach Analyzer & C2 Framework | *Python, Scikit-learn, C#*

- Developed an anomaly detection tool utilizing statistical and ML classifiers to monitor network streams and flag intrusion attempts.
- Designed a modular Command & Control (C2) proof-of-concept utilizing authenticated API-key handshakes for secure client-server communication.

EDUCATION

National Forensic Sciences University, Delhi

Bachelor of Technology in Computer Science & Engineering (Specialization: Cybersecurity) | 2024 – 2028

- **Relevant Coursework:** Network Security, Digital Forensics, DSA, Cryptography, Operating Systems, IoT

Hope Hall Foundation School, New Delhi

Senior Secondary (CBSE Class XII) — 92.8% | 2024
Secondary (CBSE Class X) — 94.6% | 2022

HONORS & ACHIEVEMENTS

- **Top 25 Rank:** Hackster CTF (College-level CTF Competition)
- **SWAYAM NPTEL Certification:** Aeromechanics of Unmanned Aerial Vehicles (Taught by Prof. Abhishek, IIT Kanpur)
- **Finalist:** Thinker's Quest Ideathon, Netaji Subhas University of Technology (NSUT), Delhi